DATA ANALYTICS PROJECT CANVAS

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Title: Detecting Scam-Induced Transactions to Mule Accounts

1. Problem Statement/Background

Mule accounts move 2-5% of total banking transactions but account for the majority of unrecovered APP fraud loss. Rule-based threshold systems fail because mule transactions sit below conventional alert ceilings and disperse within 24 hours, beyond the BoT recall window. This project builds a behaviour-based detection layer addressing the gap.

2. SMART Objectives/ Value Propositions

Value Proposition: reduce Fraud Loss Rate KPI and shorten investigation cycle from days to hours.

SMART Objectives:

- **S:** Detect mule accounts on a 2-table banking dataset using outflow velocity, counterparty diversity, off-peak frequency, and account age
- **M:** Recall ≥80%, FPR ≤5%, Precision ≥40%, detection ≤24 hrs
- **A:** Rule-based approach sufficient clear signal separation in synthetic dataset
- **R:** Supports bank's Fraud Loss Rate KPI and BoT AOC 1441 compliance
- **T:** Delivered by project deadline

3. Questions/Hypothesis

- **WHAT**: Do mule transactions have a different amount distribution from normal transactions?
  - **Hypothesis**: Scam inflows median ≥10x normal, with more round-number amounts
- **WHEN**: When during the day do scam-induced transactions concentrate?
  - **Hypothesis**: ≥50% of scam inflows in 23:00–03:00 vs ~2.6% for normal
- **WHERE**: Which channels carry mule activity, and does the mix differ between inflow and outflow legs?
  - **Hypothesis**: ≥90% digital inflows; ATM dominates outflow leg
- **WHY**: Do mule accounts move funds out faster than legitimate accounts?
  - **Hypothesis**: Mule 24h outflow ratio >0.85 vs ~0 for normal
- **HOW**: Can a 2-feature combination separate mules from legitimate high-volume accounts?
  - **Hypothesis**: Mules cluster low-diversity + high-outflow; merchants do not
- **WHO**: Are young accounts overrepresented as mules?
  - **Hypothesis**: Mule median age << normal (Mann-Whitney p<0.001)

4. Key Metrics/Attributes

- Outflow_Ratio_24h, Counterparty_Diversity, Off_Peak_Frequency, Round_Amount_Ratio, Account_Age_Days.
- **Source:** synthetic 2-table dataset (2,000 accounts + 11,985 transactions, 4% mule prevalence).

5. Analysis/Model

1. Synthesize 2-table relational data with paired scam-inflow / mule-outflow events.
2. **Clean:**
   - drop duplicates, abs() negative amounts, filter implausible amounts, impute null channels. **No outlier capping.**
3. **EDA:**
   - 6 charts (boxplot log-scale, hour heatmap, channel stack, outflow ratio histogram, diversity scatter, age-balance bubble).
4. Build Tableau dashboard with analytical charts answering 5W1H with channel, region, and temporal dimensions.

6. Findings and Insights

From the exploratory analysis across our 5W1H business questions, the following key findings and insights emerged:

1. Scam inflows median 30x normal (20,000 vs 666 THB); 76% are round-number amounts. Amount filter alone insufficient but effective as coarse pre-filter
2. 56.5% of scam inflows in 23:00–03:00 vs 2.6% normal (22x). Off-peak large transfers are high-precision triggers with low false positive risk
3. Inflows 100% digital; outflows ATM-heavy (57.6%). ATM monitoring is second line of defence if inflow alert misses
4. 85% of mules exceed 0.85 outflow ratio; median drain time = 8.1h. Strongest single-feature signal in the dataset
5. Diversity <0.6 AND outflow >0.85 uniquely isolates mules. Diversity score prevents merchant false positives
6. Mule median age = 58 days vs 1,930 days normal (33x); 75% under 90 days. First 90 days is the critical high-risk window

7. Recommendation/Action and Impact

- Deploy 4-rule cascade (Velocity, Off-Peak, NEAM, Diversity Score).
- **Action:** integrate into Fraud Investigation team's daily monitoring workflow with stepped friction rather than hard blocks.
- **Impact:** projected Recall ≥85%, FPR ≤5%, time-to-detect <24h, supporting BoT fund recall window and reducing Fraud Loss Rate KPI.